

1. Introduction

The **ShopEasy – E-Commerce Order Management Database System** is designed to manage online shopping operations through a centralized database. It stores customer, product, inventory, order, payment, shipment, and review details, improving efficiency and reducing data redundancy.

2. Business Scenario Analysis

A growing e-commerce business needs a centralized database to manage customers, products, inventory, orders, payments, and deliveries. The system replaces manual record-keeping with an efficient and organized solution.

3. Business Problem

The organization faces the following issues:

- Manual record management
 - Duplicate and inconsistent data
 - Inventory tracking difficulties
 - Slow order processing
 - Inefficient payment and shipment tracking
 - Limited business reporting
 - Increased human errors
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4. Proposed Solution

The **ShopEasy – E-Commerce Order Management Database System** provides a centralized database to:

- Manage customers and products
 - Track inventory
 - Process orders
 - Record payments
 - Monitor shipments
 - Store customer reviews
 - Generate reports
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5. Project Objectives

- Maintain customer records
 - Manage products and categories
 - Track inventory
 - Process orders efficiently
 - Record payments
 - Monitor shipments
 - Reduce manual work
 - Generate business reports
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6. Purpose of Core Entities

Entity	Purpose
Customer	Stores customer details
Category	Organizes products into categories
Product	Stores product information
Supplier	Maintains supplier details
Order	Records customer orders
Order Details	Stores ordered products and quantity
Payment	Records payment information
Shipment	Tracks delivery status
Review	Stores customer ratings and feedback

7. Major Business Processes

1. Customer Registration
 2. Product & Category Management
 3. Inventory Tracking
 4. Order Processing
 5. Payment Management
 6. Shipment Tracking
 7. Review Management
 8. Report Generation
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8. Stakeholder Analysis

Stakeholder	Responsibility
Customer	Register, order, pay, review
Administrator	Manage system operations
Supplier	Supply products
Warehouse Staff	Update inventory
Delivery Staff	Deliver orders
Finance Team	Verify payments
Management	Review reports

9. Functional Requirements

The system shall:

- Register customers
 - Manage products and categories
 - Maintain inventory
 - Process orders
 - Record payments
 - Track shipments
 - Store reviews
 - Generate reports
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10. Non-Functional Requirements

- Security
 - Reliability
 - Scalability
 - Data Integrity
 - High Performance
 - User-Friendly Interface
 - Backup & Recovery
 - Fast Response Time
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11. Expected Benefits

- Centralized data management
- Reduced redundancy
- Faster order processing
- Improved inventory control
- Better payment and shipment tracking

- Enhanced customer satisfaction
 - Accurate reporting
 - Increased efficiency
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12. Conclusion

The **ShopEasy – E-Commerce Order Management Database System** provides a centralized solution for managing e-commerce operations. It improves accuracy, simplifies business processes, reduces manual work, and supports better decision-making through efficient database management.